Module 8

Risk, Return, and the CAPM

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

Goals

- **Module 7** left two questions open,
  - What is risk?
  - How is the opportunity cost of an investment (which we use as the discount rate) related to risk?

Goals

- **Module 7** left two questions open,
  - What is risk?
  - How is the opportunity cost of an investment (which we use as the discount rate) related to risk?
- This module answers both,
  - **Part 1**, returns and risk premiums
  - **Part 2**, return volatility and portfolio diversification
  - **Part 3**, beta, the risk that actually matters
  - **Part 4**, CAPM, the relationship between beta and expected return
  - **Part 5**, from expected return to discount rate

Module 8, Part 1

Returns and Risk Premiums

Return

Recall from **Module 4**

- **Return**, percentage growth of an asset's value over some period of time

Return

Recall from **Module 4**

- **Return**, percentage growth of an asset's value over some period of time
- Suppose Oracle's stock price was \$100 at the beginning of 2024 and \$180 at the end of 2024. What was the return for 2024?

Return

Recall from **Module 4**

- **Return**, percentage growth of an asset's value over some period of time
- Suppose Oracle's stock price was \$100 at the beginning of 2024 and \$180 at the end of 2024. What was the return for 2024?

$$\text{Return} = \frac{\$180 - \$100}{\$100} = 80\%$$

Expected Returns

- You do not know what a stock will return next year

Expected Returns

- You do not know what a stock will return next year
- How do we form our expectation then?

Expected Returns

- You do not know what a stock will return next year
- How do we form our expectation then?
- We think about what might happen, the return in each case, and how likely each case is (its probability)

Expected Returns

- You do not know what a stock will return next year
- How do we form our expectation then?
- We think about what might happen, the return in each case, and how likely each case is (its probability)
- **Expected return** $E[R_i]$, what the stock returns on average across the possible cases, where likelier cases count more

Expected Returns

- You do not know what a stock will return next year
- How do we form our expectation then?
- We think about what might happen, the return in each case, and how likely each case is (its probability)
- **Expected return** $E[R_i]$, what the stock returns on average across the possible cases, where likelier cases count more
- To compute it, multiply each possible return by its probability, then add up,

$$E[R_i] = \text{Probability}_1 \times \text{Return}_1 + \text{Probability}_2 \times \text{Return}_2 + \cdots$$

Example, Expected Returns

- You want the expected return on a stock over the next year. Three things might happen,
  - A recession, 20% chance, the stock returns -10%
  - A normal year, 50% chance, the stock returns 5%
  - A boom, 30% chance, the stock returns 15%

Example, Expected Returns

- You want the expected return on a stock over the next year. Three things might happen,
  - A recession, 20% chance, the stock returns -10%
  - A normal year, 50% chance, the stock returns 5%
  - A boom, 30% chance, the stock returns 15%

- Solution, the expected return is

$$E[R_i] = 0.2 \times (-10\%) + 0.5 \times 5\% + 0.3 \times 15\% = 5\%$$

Portfolio

- **Portfolio**, the collection of securities (stocks, bonds, ...) an investor holds

Portfolio

- **Portfolio**, the collection of securities (stocks, bonds, . . .) an investor holds



WSJ, August 7, 2026, "Leopold Aschenbrenner was hailed as the 'Nostradamus of AI.' But his Situational Awareness hedge fund took on too much leverage, leading to a crash he never saw coming."

Portfolio

- **Portfolio**, the collection of securities (stocks, bonds, . . .) an investor holds



WSJ, August 7, 2026, "Leopold Aschenbrenner was hailed as the 'Nostradamus of AI.' But his Situational Awareness hedge fund took on too much leverage, leading to a crash he never saw coming."

Its stock portfolio on June 30, 2026 ($20.2 billion),

Company	Business	Weight
Sandisk	Memory chips	28%
Micron	Memory chips	28%
Bloom Energy	Fuel cells	9%
TSMC	Chip maker	6%
Nebius	AI cloud	6%
CoreWeave	AI cloud	4%
Other holdings		19%
Total		100%

Source, SEC Form 13F, filed August 14, 2026

Portfolio Returns

- **Weight** w_i of security i, the share of the portfolio's money invested in it, the weights add to 100%

Portfolio Returns

- **Weight** w_i of security i, the share of the portfolio's money invested in it, the weights add to 100%
- What is the return of a portfolio?

Portfolio Returns

- **Weight** w_i of security i, the share of the portfolio's money invested in it, the weights add to 100%
- What is the return of a portfolio?
- **Portfolio return**, the weighted average of the returns of the securities in the portfolio, for S securities,

$$R_P = w_1 \times R_1 + w_2 \times R_2 + \cdots + w_S \times R_S$$

Portfolio Returns

- **Weight** w_i of security i, the share of the portfolio's money invested in it, the weights add to 100%
- What is the return of a portfolio?
- **Portfolio return**, the weighted average of the returns of the securities in the portfolio, for S securities,

$$R_P = w_1 \times R_1 + w_2 \times R_2 + \cdots + w_S \times R_S$$

- The expected portfolio return is likewise a weighted average of the securities' expected returns,

$$E[R_P] = w_1 \times E[R_1] + w_2 \times E[R_2] + \cdots + w_S \times E[R_S]$$

Example, Portfolio Returns

- One year ago, you bought two stocks for \$500 each, \$1,000 in total
- One stock returned 30% during the year, the other -10%
- What is your portfolio return over the year?

Example, Portfolio Returns

- One year ago, you bought two stocks for \$500 each, \$1,000 in total
- One stock returned 30% during the year, the other -10%
- What is your portfolio return over the year?
- Solution, you invested $500/1{,}000 = 50\%$ in each stock, so the weights are 50% each,

$$R_P = w_1 \times R_1 + w_2 \times R_2 = 0.5 \times 30\% + 0.5 \times (-10\%) = 10\%$$

Example, Expected Portfolio Returns

- You want the expected return on your portfolio over the next year. You hold two stocks, 50% of your money in each
- The first stock's expected return is 5%, the second's is 9%
- What is your portfolio's expected return over the next year?

Example, Expected Portfolio Returns

- You want the expected return on your portfolio over the next year. You hold two stocks, 50% of your money in each
- The first stock's expected return is 5%, the second's is 9%
- What is your portfolio's expected return over the next year?

- Solution, the expected portfolio return is

$$E[R_P] = w_1 \times E[R_1] + w_2 \times E[R_2] = 0.5 \times 5\% + 0.5 \times 9\% = 7\%$$

The Risk-Free Rate and Market Returns

- $r_{f,t}$, the **risk-free rate** as of time t
  - The return from an investment with no risk, i.e., return is the same in every scenario
  - In practice, the interest rate on Treasury bills (short-term U.S. government debt)

The Risk-Free Rate and Market Returns

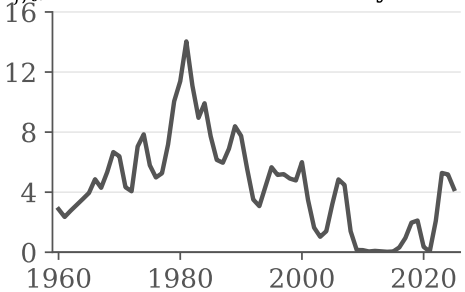
- $r_{f,t}$, the **risk-free rate** as of time t
  - The return from an investment with no risk, i.e., return is the same in every scenario
  - In practice, the interest rate on Treasury bills (short-term U.S. government debt)
- $R_{\text{mkt},t}$, the return on the market portfolio in period t
  - **Market portfolio**, the (theoretical) portfolio of all investable assets in the economy, weighted by market capitalization
  - In practice, represented by a broad market index such as the S&P 500

The Risk-Free Rate and Market Returns

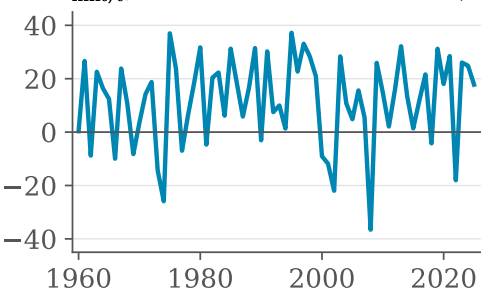
- $r_{f,t}$, the **risk-free rate** as of time t
  - The return from an investment with no risk, i.e., return is the same in every scenario
  - In practice, the interest rate on Treasury bills (short-term U.S. government debt)
- $R_{\text{mkt},t}$, the return on the market portfolio in period t
  - **Market portfolio**, the (theoretical) portfolio of all investable assets in the economy, weighted by market capitalization
  - In practice, represented by a broad market index such as the S&P 500
- $E[R_{\text{mkt}}]$, the **expected return** on the market portfolio

The Risk-Free Rate and Market Returns over Time

$r_{f,t}$, interest rate on Treasury bills (%)

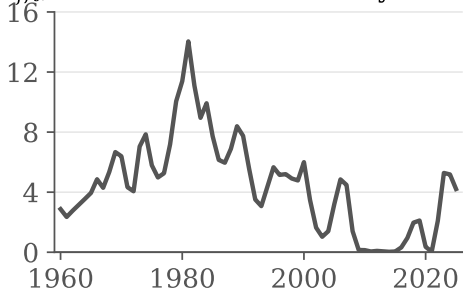


$R_{\text{mkt},t}$, return on the S&P 500 (%)

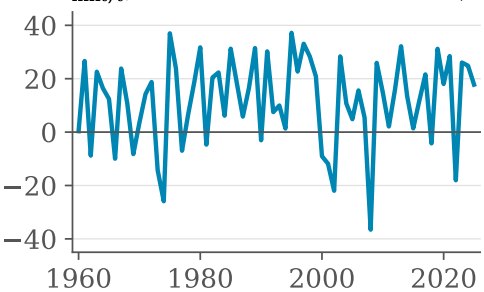


The Risk-Free Rate and Market Returns over Time

$r_{f,t}$, interest rate on Treasury bills (%)



$R_{\text{mkt},t}$, return on the S&P 500 (%)



- $r_{f,t}$ and $R_{\text{mkt},t}$ can be different at different t

Estimating Expected Returns

- Then how to estimate the expected return?

Estimating Expected Returns

- Then how to estimate the expected return?
- Past is wisdom, look at historical average returns of different portfolios,

Portfolio	Average annual return 1928–2025	Average extra return over Treasury bills
Treasury bills	3.4%	0%

Estimating Expected Returns

- Then how to estimate the expected return?
- Past is wisdom, look at historical average returns of different portfolios,

Portfolio	Average annual return 1928–2025	Average extra return over Treasury bills
Treasury bills	3.4%	0%
Investment-grade corporate bonds	6.9%	3.5%

Estimating Expected Returns

- Then how to estimate the expected return?
- Past is wisdom, look at historical average returns of different portfolios,

Portfolio	Average annual return 1928–2025	Average extra return over Treasury bills
Treasury bills	3.4%	0%
Investment-grade corporate bonds	6.9%	3.5%
Stocks	11.9%	8.5%

Estimating Expected Returns

- Then how to estimate the expected return?
- Past is wisdom, look at historical average returns of different portfolios,

Portfolio	Average annual return 1928–2025	Average extra return over Treasury bills
Treasury bills	3.4%	0%
Investment-grade corporate bonds	6.9%	3.5%
Stocks	11.9%	8.5%

- Stocks are riskier than bonds, which are riskier than Treasury bills

Estimating Expected Returns

- Then how to estimate the expected return?
- Past is wisdom, look at historical average returns of different portfolios,

Portfolio	Average annual return 1928–2025	Average extra return over Treasury bills
Treasury bills	3.4%	0%
Investment-grade corporate bonds	6.9%	3.5%
Stocks	11.9%	8.5%

- Stocks are riskier than bonds, which are riskier than Treasury bills
- Naturally, we expect riskier assets to have higher expected returns

Risk Premiums

- **Risk premium**, the "extra" expected return you earn on an investment over the risk-free rate,

$$E[R_i] = r_f + \text{Risk premium}_i$$

Risk Premiums

- **Risk premium**, the "extra" expected return you earn on an investment over the risk-free rate,

$$E[R_i] = r_f + \text{Risk premium}_i$$

- The average extra return over Treasury bills is our estimate of the risk premium

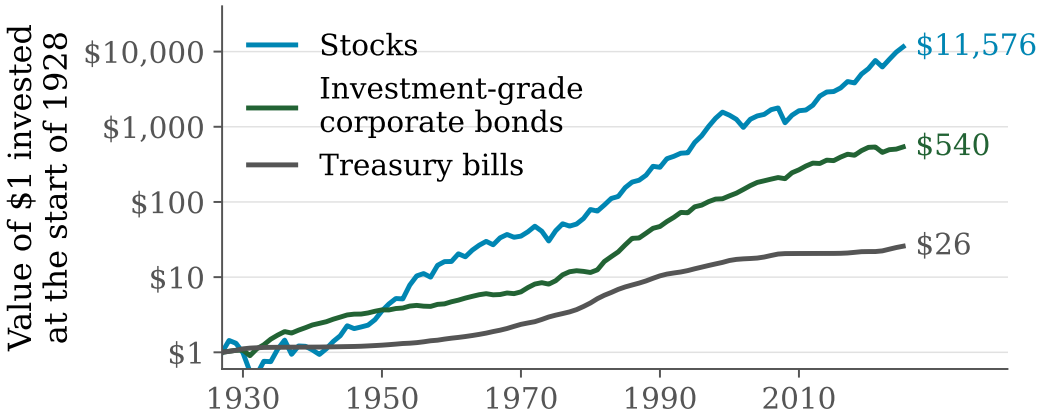
Risk Premiums

- **Risk premium**, the "extra" expected return you earn on an investment over the risk-free rate,

$$E[R_i] = r_f + \text{Risk premium}_i$$

- The average extra return over Treasury bills is our estimate of the risk premium
- For investment-grade corporate bonds that is 3.5%, for stocks 8.5%

Returns over Time for Stocks, Bonds, and Treasury Bills



Recap of Part 1

- **Return**, percentage growth of an asset's value over some period of time

Recap of Part 1

- **Return**, percentage growth of an asset's value over some period of time
- **Expected return**, multiply each possible return by its probability, then add up

Recap of Part 1

- **Return**, percentage growth of an asset's value over some period of time
- **Expected return**, multiply each possible return by its probability, then add up
- **Portfolio return**, the weighted average of the securities' returns

Recap of Part 1

- **Return**, percentage growth of an asset's value over some period of time
- **Expected return**, multiply each possible return by its probability, then add up
- **Portfolio return**, the weighted average of the securities' returns
- $E[R_i] = r_f + \text{Risk premium}_i$, riskier assets have higher expected returns

Recap of Part 1

- **Return**, percentage growth of an asset's value over some period of time
- **Expected return**, multiply each possible return by its probability, then add up
- **Portfolio return**, the weighted average of the securities' returns
- $E[R_i] = r_f + \text{Risk premium}_i$, riskier assets have higher expected returns
- The risk premium of stocks is about 8.5%

Module 8, Part 2

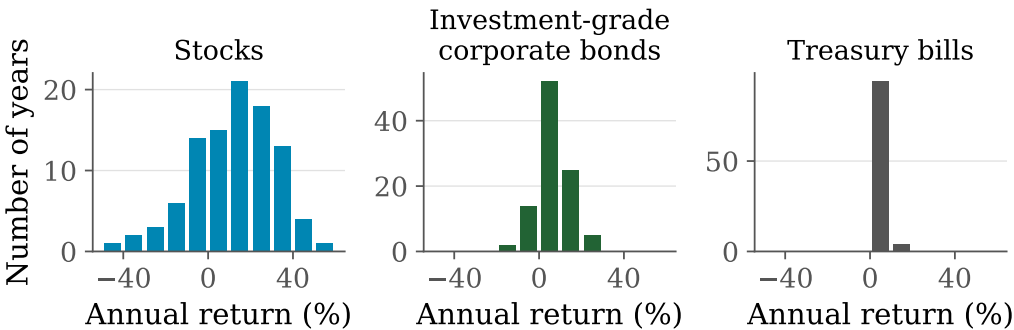
Return Volatility and Portfolio Diversification

The Risk Premium of Stocks Is High, but What about Volatility?

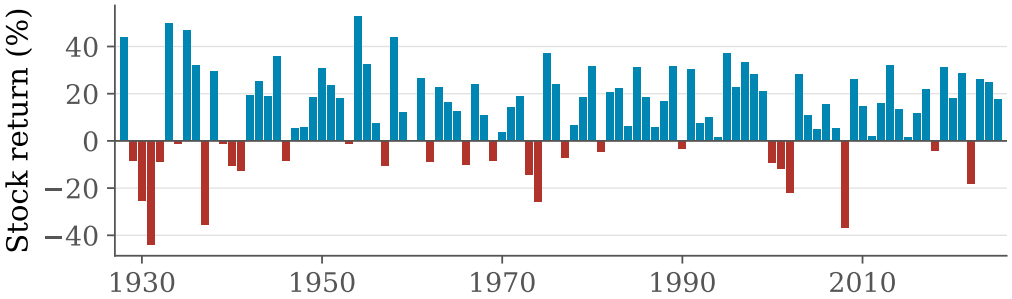
- How volatile are portfolios of stocks vs. bonds vs. Treasury bills?

The Risk Premium of Stocks Is High, but What about Volatility?

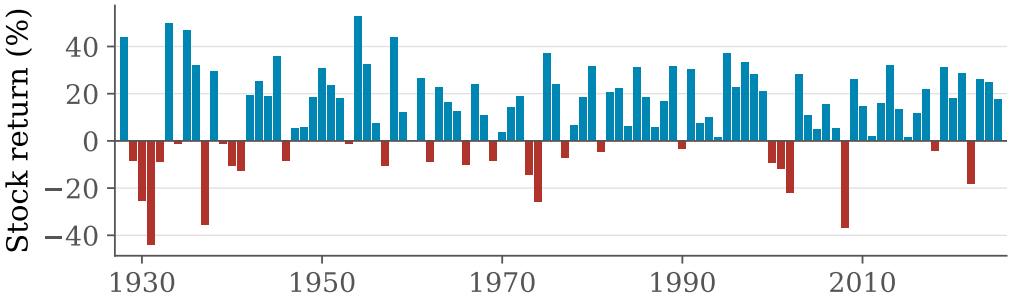
- How volatile are portfolios of stocks vs. bonds vs. Treasury bills?
- Consider the variation in annual returns, 1928–2025, how many years fell in each range,



Stock Market Returns over Time

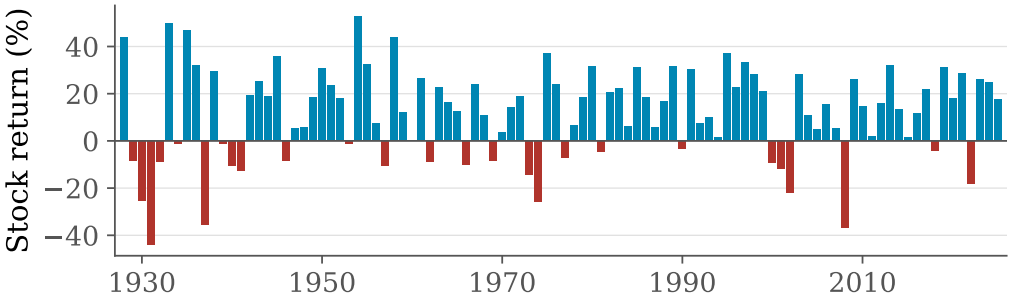


Stock Market Returns over Time



- A portfolio of all stocks has had negative returns in around 25% of all years

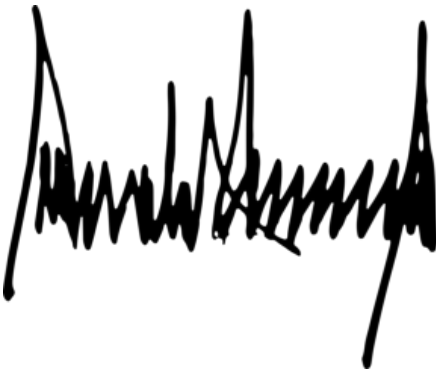
Stock Market Returns over Time



- A portfolio of all stocks has had negative returns in around 25% of all years
- Individual stocks are even more volatile, around 50% of stock-years have negative returns

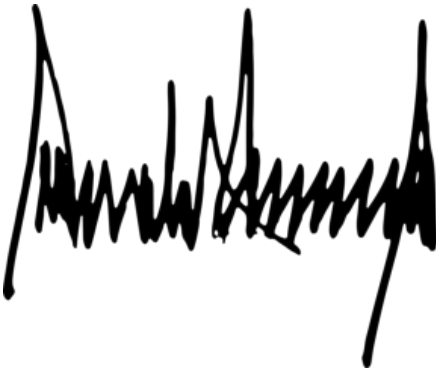
Discuss, Do They Look Alike?

US President Donald Trump's Signature

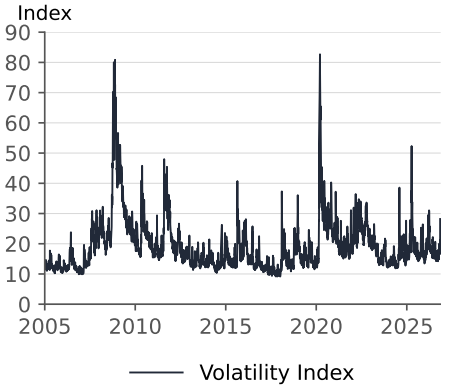


Discuss, Do They Look Alike?

US President Donald Trump's Signature



US Stock Volatility Index (VIX)



Measuring Volatility, Variance and Standard Deviation

- **Variance**, the average squared deviation from the average return, in four steps,
  1. Compute the average return

Measuring Volatility, Variance and Standard Deviation

- **Variance**, the average squared deviation from the average return, in four steps,
  1. Compute the average return
  2. Take each return's deviation from the average

Measuring Volatility, Variance and Standard Deviation

- **Variance**, the average squared deviation from the average return, in four steps,
  1. Compute the average return
  2. Take each return's deviation from the average
  3. Square each deviation

Measuring Volatility, Variance and Standard Deviation

- **Variance**, the average squared deviation from the average return, in four steps,
  1. Compute the average return
  2. Take each return's deviation from the average
  3. Square each deviation
  4. Average the squared deviations, that is the variance

Measuring Volatility, Variance and Standard Deviation

- **Variance**, the average squared deviation from the average return, in four steps,
  1. Compute the average return
  2. Take each return's deviation from the average
  3. Square each deviation
  4. Average the squared deviations, that is the variance

- **Standard deviation**, the square root of the variance,

$$\text{Standard deviation} = \sqrt{\text{Variance}}$$

Measuring Volatility, Variance and Standard Deviation

- Your returns over six years are 5%, 25%, 10%, −5%, 5%, and 20%. What is the standard deviation of returns?

Measuring Volatility, Variance and Standard Deviation

- Your returns over six years are 5%, 25%, 10%, −5%, 5%, and 20%. What is the standard deviation of returns?

Year	Return
1	5%
2	25%
3	10%
4	−5%
5	5%
6	20%

Measuring Volatility, Variance and Standard Deviation

- Your returns over six years are 5%, 25%, 10%, −5%, 5%, and 20%. What is the standard deviation of returns?

Year	Return
1	5%
2	25%
3	10%
4	−5%
5	5%
6	20%
Average	10%

Measuring Volatility, Variance and Standard Deviation

- Your returns over six years are 5%, 25%, 10%, −5%, 5%, and 20%. What is the standard deviation of returns?

Year	Return	Deviation from average
1	5%	−5%
2	25%	15%
3	10%	0%
4	−5%	−15%
5	5%	−5%
6	20%	10%
Average	10%	

Measuring Volatility, Variance and Standard Deviation

- Your returns over six years are 5%, 25%, 10%, −5%, 5%, and 20%. What is the standard deviation of returns?

Year	Return	Deviation from average	Squared deviation
1	5%	−5%	0.0025
2	25%	15%	0.0225
3	10%	0%	0.0000
4	−5%	−15%	0.0225
5	5%	−5%	0.0025
6	20%	10%	0.0100
Average	10%		

Measuring Volatility, Variance and Standard Deviation

- Your returns over six years are 5%, 25%, 10%, −5%, 5%, and 20%. What is the standard deviation of returns?

Year	Return	Deviation from average	Squared deviation
1	5%	−5%	0.0025
2	25%	15%	0.0225
3	10%	0%	0.0000
4	−5%	−15%	0.0225
5	5%	−5%	0.0025
6	20%	10%	0.0100
Average	10%	Variance (average of squared deviations)	0.0100

Measuring Volatility, Variance and Standard Deviation

- Your returns over six years are 5%, 25%, 10%, −5%, 5%, and 20%. What is the standard deviation of returns?

Year	Return	Deviation from average	Squared deviation
1	5%	−5%	0.0025
2	25%	15%	0.0225
3	10%	0%	0.0000
4	−5%	−15%	0.0225
5	5%	−5%	0.0025
6	20%	10%	0.0100
Average	10%	Variance (average of squared deviations)	0.0100
		Standard deviation (square root)	10%

Standard Deviation of Treasury Bills vs. Bonds vs. Stocks

Portfolio	Average annual return 1928–2025	Standard deviation 1928–2025
Treasury bills	3.4%	3.0%

Standard Deviation of Treasury Bills vs. Bonds vs. Stocks

Portfolio	Average annual return 1928–2025	Standard deviation 1928–2025
Treasury bills	3.4%	3.0%
Investment-grade corporate bonds	6.9%	7.7%

Standard Deviation of Treasury Bills vs. Bonds vs. Stocks

Portfolio	Average annual return 1928–2025	Standard deviation 1928–2025
Treasury bills	3.4%	3.0%
Investment-grade corporate bonds	6.9%	7.7%
Stocks	11.9%	19.4%

Measuring Volatility, Individual Stocks

- The standard deviation of an individual stock is typically much higher than that of the index

Measuring Volatility, Individual Stocks

- The standard deviation of an individual stock is typically much higher than that of the index

Stock	Standard deviation of annual returns (%)
Tesla	201.4
Amazon	186.9
Apple	65.1
Microsoft	41.9
McDonald's	41.3

Measuring Volatility, Individual Stocks

- The standard deviation of an individual stock is typically much higher than that of the index

Stock	Standard deviation of annual returns (%)
Tesla	201.4
Amazon	186.9
Apple	65.1
Microsoft	41.9
McDonald's	41.3
S&P 500	**19.4**

Measuring Volatility, Individual Stocks

- The standard deviation of an individual stock is typically much higher than that of the index
- This is the effect of diversification

Stock	Standard deviation of annual returns (%)
Tesla	201.4
Amazon	186.9
Apple	65.1
Microsoft	41.9
McDonald's	41.3
S&P 500	**19.4**

Example, Portfolio Diversification

- You consider two stocks, stock A and stock B

Example, Portfolio Diversification

- You consider two stocks, stock A and stock B
- Each has a good year or a bad year next year, depending on its own events (say, its product succeeds or flops)

Example, Portfolio Diversification

- You consider two stocks, stock A and stock B
- Each has a good year or a bad year next year, depending on its own events (say, its product succeeds or flops)
- One of four scenarios happens with equal likelihood,

Scenario	Probability	Stock A (%)	Stock B (%)
A good, B good	1/4	+25	+20
A good, B bad	1/4	+25	−10
A bad, B good	1/4	−5	+20
A bad, B bad	1/4	−5	−10

Example, Portfolio Diversification

- As investors, we love expected return but hate volatility (standard deviation)

Example, Portfolio Diversification

- As investors, we love expected return but hate volatility (standard deviation)
- So we want a high expected return for a low standard deviation

Example, Portfolio Diversification

- As investors, we love expected return but hate volatility (standard deviation)
- So we want a high expected return for a low standard deviation
- Then how should we invest?

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return
Good year	25%
Bad year	−5%

Stock B	Return
Good year	20%
Bad year	−10%

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return
Good year	25%
Bad year	−5%
$E[R_A]$	10%

Stock B	Return
Good year	20%
Bad year	−10%
$E[R_B]$	5%

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return	Deviation from $E[R_A]$	
Good year	25%	15%	
Bad year	-5%	-15%	
$E[R_A]$	10%		

Stock B	Return	Deviation from $E[R_B]$	
Good year	20%	15%	
Bad year	-10%	-15%	
$E[R_B]$	5%		

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return	Deviation from $E[R_A]$	Squared deviation
Good year	25%	15%	0.0225
Bad year	-5%	-15%	0.0225
$E[R_A]$	10%		

Stock B	Return	Deviation from $E[R_B]$	Squared deviation
Good year	20%	15%	0.0225
Bad year	-10%	-15%	0.0225
$E[R_B]$	5%		

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return	Deviation from $E[R_A]$	Squared deviation
Good year	25%	15%	0.0225
Bad year	-5%	-15%	0.0225
$E[R_A]$	10%	Variance	0.0225

Stock B	Return	Deviation from $E[R_B]$	Squared deviation
Good year	20%	15%	0.0225
Bad year	-10%	-15%	0.0225
$E[R_B]$	5%	Variance	0.0225

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return	Deviation from $E[R_A]$	Squared deviation
Good year	25%	15%	0.0225
Bad year	−5%	−15%	0.0225
$E[R_A]$	10%	Variance	0.0225
		Std. dev.	15.0%

Stock B	Return	Deviation from $E[R_B]$	Squared deviation
Good year	20%	15%	0.0225
Bad year	−10%	−15%	0.0225
$E[R_B]$	5%	Variance	0.0225
		Std. dev.	15.0%

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return	Deviation from $E[R_A]$	Squared deviation
Good year	25%	15%	0.0225
Bad year	−5%	−15%	0.0225
$E[R_A]$	10%	Variance	0.0225
		Std. dev.	15.0%

Stock B	Return	Deviation from $E[R_B]$	Squared deviation
Good year	20%	15%	0.0225
Bad year	−10%	−15%	0.0225
$E[R_B]$	5%	Variance	0.0225
		Std. dev.	15.0%

- $E[R]$ / std. dev., stock A $10/15 = 0.67$, stock B $5/15 = 0.33$

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return	Deviation from $E[R_A]$	Squared deviation
Good year	25%	15%	0.0225
Bad year	−5%	−15%	0.0225
$E[R_A]$	10%	Variance	0.0225
		Std. dev.	15.0%

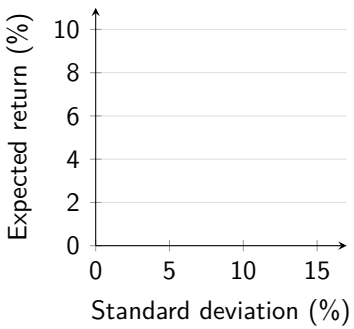
Stock B	Return	Deviation from $E[R_B]$	Squared deviation
Good year	20%	15%	0.0225
Bad year	−10%	−15%	0.0225
$E[R_B]$	5%	Variance	0.0225
		Std. dev.	15.0%

- $E[R]$ / std. dev., stock A $10/15 = 0.67$, stock B $5/15 = 0.33$
- Stock A looks better as a stand-alone investment on this metric

Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

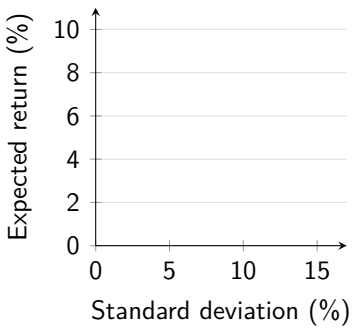
Portfolio	
Stock A weight	
Stock B weight	
A good, B good	
A good, B bad	
A bad, B good	
A bad, B bad	
$E[R]$	
Std. dev.	
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

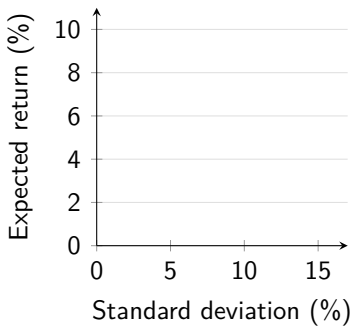
Portfolio	1
Stock A weight	100%
Stock B weight	
A good, B good	
A good, B bad	
A bad, B good	
A bad, B bad	
$E[R]$	
Std. dev.	
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

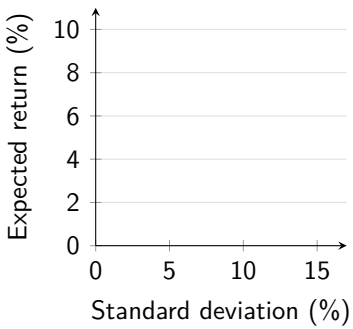
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	
A good, B bad	
A bad, B good	
A bad, B bad	
$E[R]$	
Std. dev.	
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

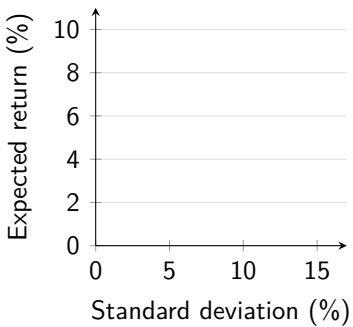
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	
A bad, B good	
A bad, B bad	
$E[R]$	
Std. dev.	
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

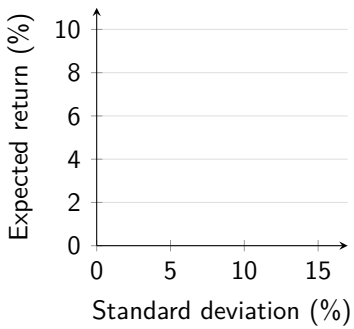
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	
A bad, B bad	
$E[R]$	
Std. dev.	
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

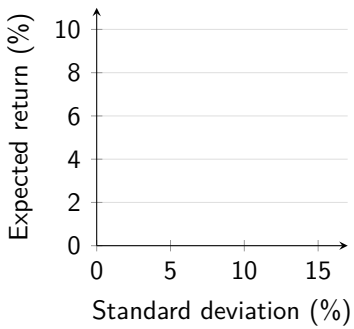
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	−5%
A bad, B bad	
$E[R]$	
Std. dev.	
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

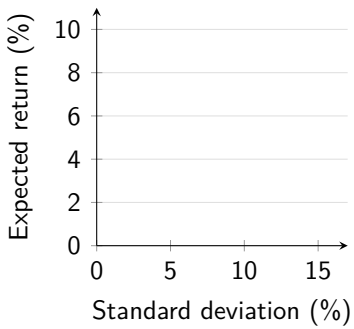
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	−5%
A bad, B bad	−5%
$E[R]$	
Std. dev.	
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

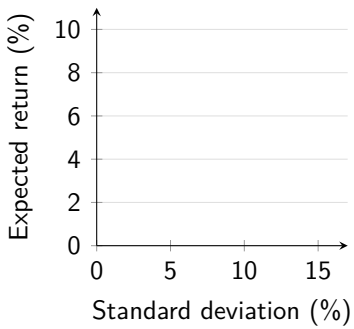
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	−5%
A bad, B bad	−5%
$E[R]$	10.0%
Std. dev.	
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

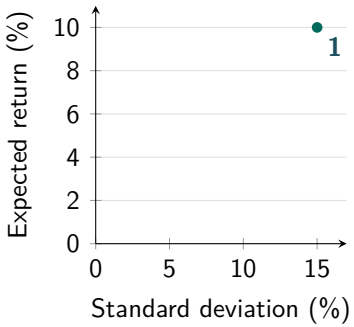
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	−5%
A bad, B bad	−5%
$E[R]$	10.0%
Std. dev.	15.0%
$E[R]$ / std. dev.	



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

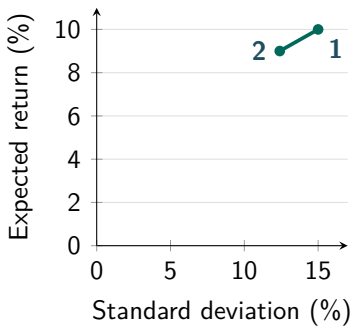
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	−5%
A bad, B bad	−5%
$E[R]$	10.0%
Std. dev.	15.0%
$E[R]$ / std. dev.	0.67



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

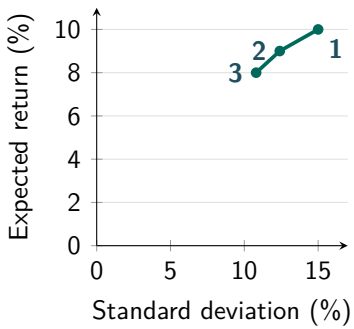
Portfolio	1	2
Stock A weight	100%	80%
Stock B weight	0%	20%
A good, B good	25%	24%
A good, B bad	25%	18%
A bad, B good	−5%	0%
A bad, B bad	−5%	−6%
$E[R]$	10.0%	9.0%
Std. dev.	15.0%	12.4%
$E[R]$ / std. dev.	0.67	0.73



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

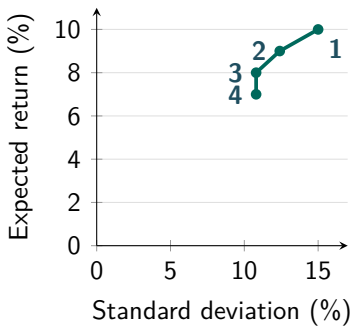
Portfolio	1	2	3
Stock A weight	100%	80%	60%
Stock B weight	0%	20%	40%
A good, B good	25%	24%	23%
A good, B bad	25%	18%	11%
A bad, B good	−5%	0%	5%
A bad, B bad	−5%	−6%	−7%
$E[R]$	10.0%	9.0%	8.0%
Std. dev.	15.0%	12.4%	10.8%
$E[R]$ / std. dev.	0.67	0.73	0.74



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

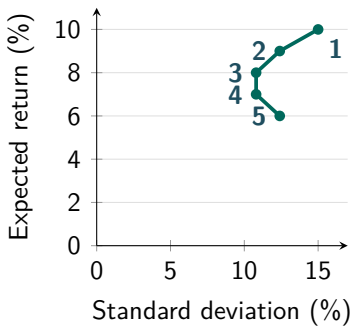
Portfolio	1	2	3	4
Stock A weight	100%	80%	60%	40%
Stock B weight	0%	20%	40%	60%
A good, B good	25%	24%	23%	22%
A good, B bad	25%	18%	11%	4%
A bad, B good	−5%	0%	5%	10%
A bad, B bad	−5%	−6%	−7%	−8%
$E[R]$	10.0%	9.0%	8.0%	7.0%
Std. dev.	15.0%	12.4%	10.8%	10.8%
$E[R]$ / std. dev.	0.67	0.73	0.74	0.65



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

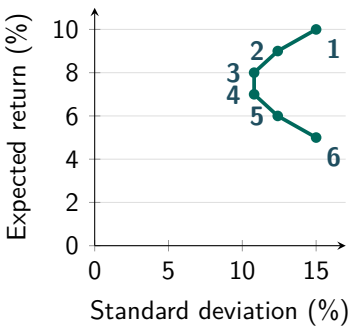
Portfolio	1	2	3	4	5
Stock A weight	100%	80%	60%	40%	20%
Stock B weight	0%	20%	40%	60%	80%
A good, B good	25%	24%	23%	22%	21%
A good, B bad	25%	18%	11%	4%	−3%
A bad, B good	−5%	0%	5%	10%	15%
A bad, B bad	−5%	−6%	−7%	−8%	−9%
$E[R]$	10.0%	9.0%	8.0%	7.0%	6.0%
Std. dev.	15.0%	12.4%	10.8%	10.8%	12.4%
$E[R]$ / std. dev.	0.67	0.73	0.74	0.65	0.49



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

Portfolio	1	2	3	4	5	6
Stock A weight	100%	80%	60%	40%	20%	0%
Stock B weight	0%	20%	40%	60%	80%	100%
A good, B good	25%	24%	23%	22%	21%	20%
A good, B bad	25%	18%	11%	4%	−3%	−10%
A bad, B good	−5%	0%	5%	10%	15%	20%
A bad, B bad	−5%	−6%	−7%	−8%	−9%	−10%
$E[R]$	10.0%	9.0%	8.0%	7.0%	6.0%	5.0%
Std. dev.	15.0%	12.4%	10.8%	10.8%	12.4%	15.0%
$E[R]$ / std. dev.	0.67	0.73	0.74	0.65	0.49	0.33



Portfolio Diversification

- What is the standard deviation of the portfolio?

$$\text{StdDev}[R_P] \neq w_A \,\text{StdDev}[R_A] + w_B \,\text{StdDev}[R_B]$$

Portfolio Diversification

- What is the standard deviation of the portfolio?

$$\text{StdDev}[R_P] \neq w_A \,\text{StdDev}[R_A] + w_B \,\text{StdDev}[R_B]$$

- E.g., portfolio 3 (60% stock A, 40% stock B), $0.6 \times 15\% + 0.4 \times 15\% = 15\%$, but its standard deviation is only 10.8%

Portfolio Diversification

- What is the standard deviation of the portfolio?

$$\text{StdDev}[R_P] \neq w_A \,\text{StdDev}[R_A] + w_B \,\text{StdDev}[R_B]$$

- E.g., portfolio 3 (60% stock A, 40% stock B), $0.6 \times 15\% + 0.4 \times 15\% = 15\%$, but its standard deviation is only 10.8%
- **Diversification**, spreading the money in a portfolio across many securities rather than one

Portfolio Diversification

- What is the standard deviation of the portfolio?

$$\text{StdDev}[R_P] \neq w_A \,\text{StdDev}[R_A] + w_B \,\text{StdDev}[R_B]$$

- E.g., portfolio 3 (60% stock A, 40% stock B), $0.6 \times 15\% + 0.4 \times 15\% = 15\%$, but its standard deviation is only 10.8%
- **Diversification**, spreading the money in a portfolio across many securities rather than one
- Diversification typically makes the standard deviation of a portfolio lower than the weighted average of the standard deviations of the stocks in it,

$$\textbf{StdDev}[R_P] < w_A \,\text{StdDev}[R_A] + w_B \,\text{StdDev}[R_B]$$

Simulation with AI

- An economy with N companies, each producing a product

Simulation with AI

- An economy with N companies, each producing a product
- Each company's stock returns 20% in a normal year

Simulation with AI

- An economy with N companies, each producing a product
- Each company's stock returns 20% in a normal year
- Two kinds of risk,
  - A company's product succeeds (30% chance each year), its stock return rises by 40 points
  - A company's product fails (30% chance each year), its stock return falls by 40 points

Simulation with AI

- An economy with N companies, each producing a product
- Each company's stock returns 20% in a normal year
- Two kinds of risk,
  - A company's product succeeds (30% chance each year), its stock return rises by 40 points
  - A company's product fails (30% chance each year), its stock return falls by 40 points
  - Something bad hits the whole economy, like a pandemic (10% chance each year), every stock return falls by 30 points

Simulating an Investor's Return

- Simulate many years of this economy, the AI writes the code and plots

Simulating an Investor's Return

- Simulate many years of this economy, the AI writes the code and plots
- An investor puts all the money in one company. What is the standard deviation of the investor's return?

Simulating an Investor's Return

- Simulate many years of this economy, the AI writes the code and plots
- An investor puts all the money in one company. What is the standard deviation of the investor's return?
- Now the investor splits the money equally across 2 companies, then 5, 10, 30, 100. Does the standard deviation fall? To zero?

What Is the End Game of Portfolio Diversification?

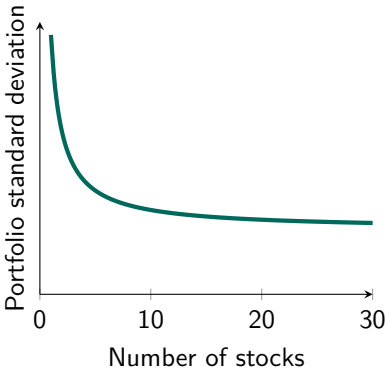
- Stocks do not move perfectly together

What Is the End Game of Portfolio Diversification?

- Stocks do not move perfectly together
- So a portfolio's standard deviation is below the average of its stocks'

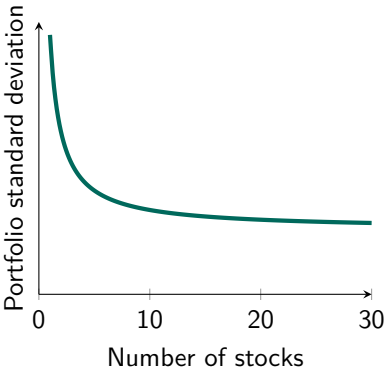
What Is the End Game of Portfolio Diversification?

- Stocks do not move perfectly together
- So a portfolio's standard deviation is below the average of its stocks'
- The more stocks, the lower it goes



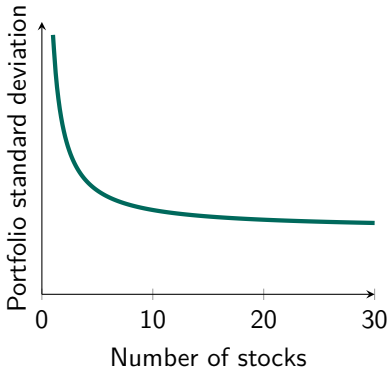
What Is the End Game of Portfolio Diversification?

- Stocks do not move perfectly together
- So a portfolio's standard deviation is below the average of its stocks'
- The more stocks, the lower it goes
- **Firm-specific risk**, the risk from events that hit one company (a product flop, a lawsuit)



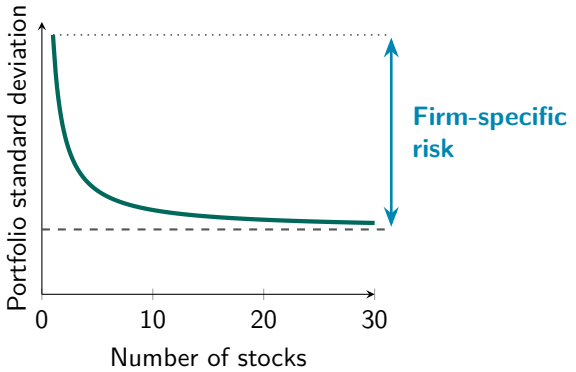
What Is the End Game of Portfolio Diversification?

- Stocks do not move perfectly together
- So a portfolio's standard deviation is below the average of its stocks'
- The more stocks, the lower it goes
- **Firm-specific risk**, the risk from events that hit one company (a product flop, a lawsuit)
- Such events average out across many stocks



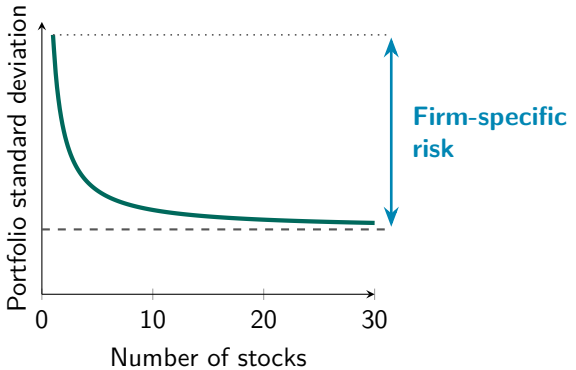
What Is the End Game of Portfolio Diversification?

- Stocks do not move perfectly together
- So a portfolio's standard deviation is below the average of its stocks'
- The more stocks, the lower it goes
- **Firm-specific risk**, the risk from events that hit one company (a product flop, a lawsuit)
- Such events average out across many stocks
- Diversification removes firm-specific risk



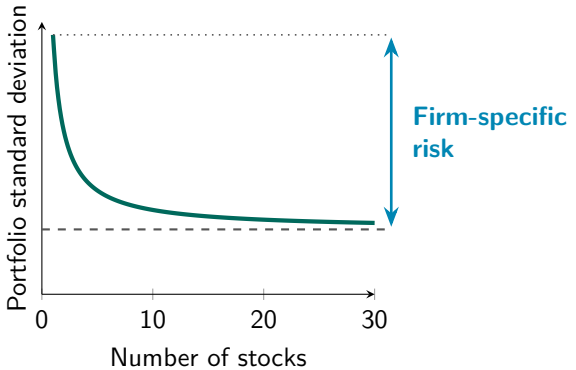
What Is the End Game of Portfolio Diversification?

- Yet there is a limit, the curve never reaches zero



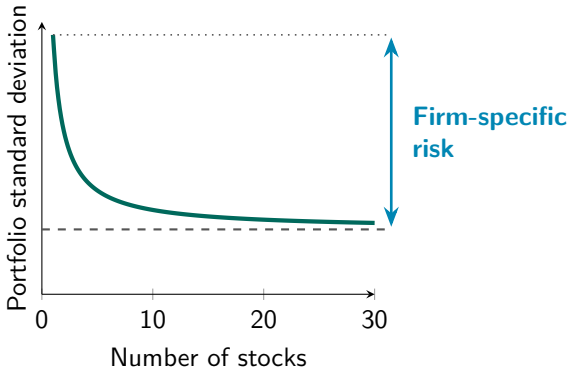
What Is the End Game of Portfolio Diversification?

- Yet there is a limit, the curve never reaches zero
- **Systematic risk**, the risk from economy-wide events that move all stocks together (a pandemic)



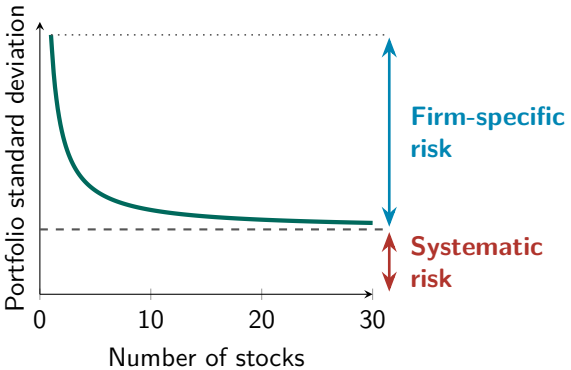
What Is the End Game of Portfolio Diversification?

- Yet there is a limit, the curve never reaches zero
- **Systematic risk**, the risk from economy-wide events that move all stocks together (a pandemic)
- Every stock is hit at once, nothing averages out



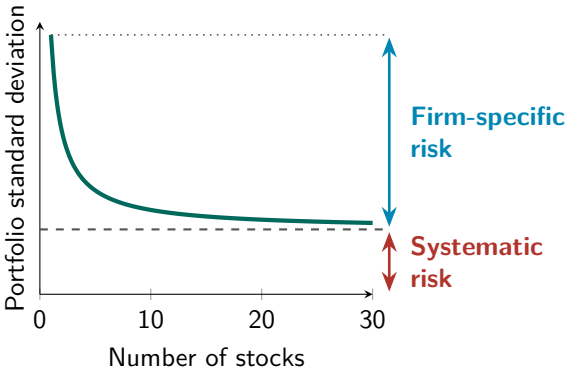
What Is the End Game of Portfolio Diversification?

- Yet there is a limit, the curve never reaches zero
- **Systematic risk**, the risk from economy-wide events that move all stocks together (a pandemic)
- Every stock is hit at once, nothing averages out
- Diversification cannot remove systematic risk



What Is the End Game of Portfolio Diversification?

- Yet there is a limit, the curve never reaches zero
- **Systematic risk**, the risk from economy-wide events that move all stocks together (a pandemic)
- Every stock is hit at once, nothing averages out
- Diversification cannot remove systematic risk



The end game, a diversified portfolio carries systematic risk only

Recap of Part 2

- **Volatility**, how much returns vary, the **variance** or **standard deviation**

Recap of Part 2

- **Volatility**, how much returns vary, the **variance** or **standard deviation**
- Stocks vary far more than bonds and Treasury bills, individual stocks even more

Recap of Part 2

- **Volatility**, how much returns vary, the **variance** or **standard deviation**
- Stocks vary far more than bonds and Treasury bills, individual stocks even more
- **Diversification**, a portfolio's standard deviation is below its stocks' average

Recap of Part 2

- **Volatility**, how much returns vary, the **variance** or **standard deviation**
- Stocks vary far more than bonds and Treasury bills, individual stocks even more
- **Diversification**, a portfolio's standard deviation is below its stocks' average
- Diversification removes firm-specific risk, not systematic risk

Module 8, Part 3

Beta

From Standard Deviation to Beta

- **Part 2**'s lesson, a diversified investor is left with systematic risk only

From Standard Deviation to Beta

- **Part 2**'s lesson, a diversified investor is left with systematic risk only
- So the risk of a stock that matters is its systematic risk, not its standard deviation

From Standard Deviation to Beta

- **Part 2**'s lesson, a diversified investor is left with systematic risk only
- So the risk of a stock that matters is its systematic risk, not its standard deviation
- How much systematic risk does a stock carry?

The Market Portfolio

- Recall the **market portfolio**, the (theoretical) portfolio of all investable assets in the economy, weighted by market capitalization

The Market Portfolio

- Recall the **market portfolio**, the (theoretical) portfolio of all investable assets in the economy, weighted by market capitalization
- It is *fully* diversified and has no firm-specific risk

The Market Portfolio

- Recall the **market portfolio**, the (theoretical) portfolio of all investable assets in the economy, weighted by market capitalization
- It is *fully* diversified and has no firm-specific risk
- It only has systematic risk (e.g., a government shutdown, a pandemic)

The Market Portfolio

- Recall the **market portfolio**, the (theoretical) portfolio of all investable assets in the economy, weighted by market capitalization
- It is *fully* diversified and has no firm-specific risk
- It only has systematic risk (e.g., a government shutdown, a pandemic)
- In practice, represented by a broad market index such as the S&P 500

Beta

- Recall the **return**, percentage growth of an asset's value over some period of time

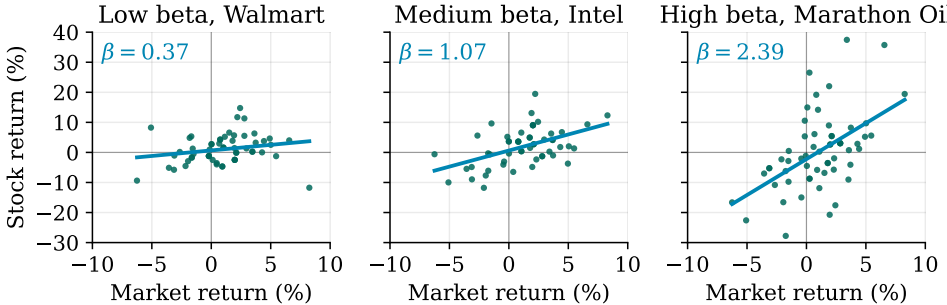
Beta

- Recall the **return**, percentage growth of an asset's value over some period of time
- **Excess return**, a security's realized return over the risk-free rate in period t, $R_{i,t} - r_{f,t}$

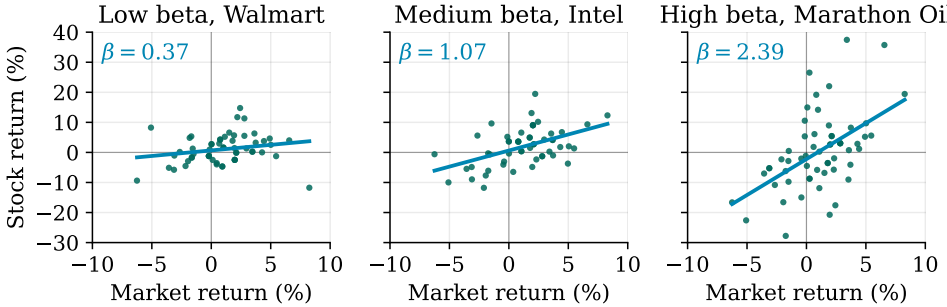
Beta

- Recall the **return**, percentage growth of an asset's value over some period of time
- **Excess return**, a security's realized return over the risk-free rate in period t, $R_{i,t} - r_{f,t}$
- **Beta**, the sensitivity of a security's (or portfolio's) excess return to the market portfolio's excess return

Example, Low vs. Medium vs. High Beta Stocks

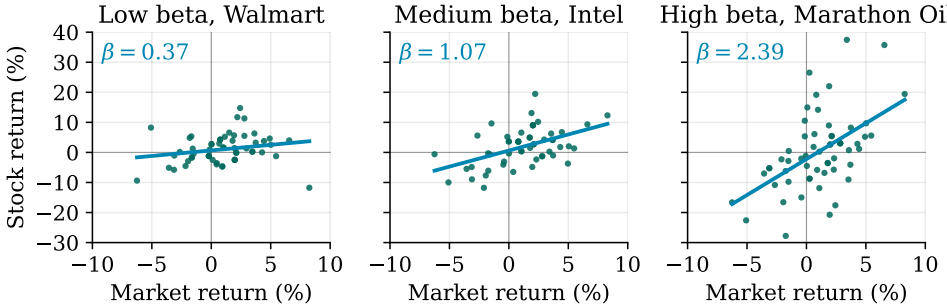


Example, Low vs. Medium vs. High Beta Stocks



- Cyclical businesses, whose revenues and earnings depend strongly on the state of the economy, tend to have high betas

Example, Low vs. Medium vs. High Beta Stocks



- Cyclical businesses, whose revenues and earnings depend strongly on the state of the economy, tend to have high betas
- Businesses that produce essentials (food, beer, cosmetics) are less affected by the state of the economy and tend to have low betas

Portfolio Beta

- Now suppose a portfolio holds two or more securities. What is its beta?

Portfolio Beta

- Now suppose a portfolio holds two or more securities. What is its beta?
- **Portfolio beta**, the weighted average of the betas of the securities in the portfolio, with the same weights w_i as for portfolio returns,

$$\beta_P = w_1 \times \beta_1 + w_2 \times \beta_2 + \cdots$$

Portfolio Beta

- Now suppose a portfolio holds two or more securities. What is its beta?
- **Portfolio beta**, the weighted average of the betas of the securities in the portfolio, with the same weights w_i as for portfolio returns,

$$\beta_P = w_1 \times \beta_1 + w_2 \times \beta_2 + \cdots$$

- Diversification lowers a portfolio's standard deviation, but it does ***not*** lower beta

Portfolio Beta

- Now suppose a portfolio holds two or more securities. What is its beta?
- **Portfolio beta**, the weighted average of the betas of the securities in the portfolio, with the same weights w_i as for portfolio returns,

$$\beta_P = w_1 \times \beta_1 + w_2 \times \beta_2 + \cdots$$

- Diversification lowers a portfolio's standard deviation, but it does ***not*** lower beta
- Beta measures systematic risk, which does not go away with diversification

Portfolio Beta

- Now suppose a portfolio holds two or more securities. What is its beta?
- **Portfolio beta**, the weighted average of the betas of the securities in the portfolio, with the same weights w_i as for portfolio returns,

$$\beta_P = w_1 \times \beta_1 + w_2 \times \beta_2 + \cdots$$

- Diversification lowers a portfolio's standard deviation, but it does ***not*** lower beta
- Beta measures systematic risk, which does not go away with diversification
- The market portfolio itself has a beta of 1

Portfolio Beta

- Now suppose a portfolio holds two or more securities. What is its beta?
- **Portfolio beta**, the weighted average of the betas of the securities in the portfolio, with the same weights w_i as for portfolio returns,

$$\beta_P = w_1 \times \beta_1 + w_2 \times \beta_2 + \cdots$$

- Diversification lowers a portfolio's standard deviation, but it does ***not*** lower beta
- Beta measures systematic risk, which does not go away with diversification
- The market portfolio itself has a beta of 1
  - Its return moves one for one with the market's, because it is the market

A Primer on Regression Analysis

- How do we estimate beta?

A Primer on Regression Analysis

- How do we estimate beta?
- **Regression analysis**, a statistical tool for describing the "best-fit" linear relation between two variables, X (the "explanatory" variable) and Y (the "dependent" variable)

What the Regression Does

- We have data points, i.e., pairs of x and y

What the Regression Does

- We have data points, i.e., pairs of x and y
- We also have an equation, where i indexes data points,

$$y_i = \alpha + \beta \times x_i + e_i$$

  - α is the "intercept", β is the "coefficient" on X, and the e_i are the "residuals"

What the Regression Does

- We have data points, i.e., pairs of x and y
- We also have an equation, where i indexes data points,

$$y_i = \alpha + \beta \times x_i + e_i$$

  - α is the "intercept", β is the "coefficient" on X, and the e_i are the "residuals"
- A regression graphically finds the best-fit line between the two variables, α is the y-intercept and β the slope

What the Regression Does

- We have data points, i.e., pairs of x and y
- We also have an equation, where i indexes data points,

$$y_i = \alpha + \beta \times x_i + e_i$$

  - α is the "intercept", β is the "coefficient" on X, and the e_i are the "residuals"

- A regression graphically finds the best-fit line between the two variables, α is the y-intercept and β the slope
- The best-fit line keeps the residuals e_i as small as possible overall

Example, Height and Weight

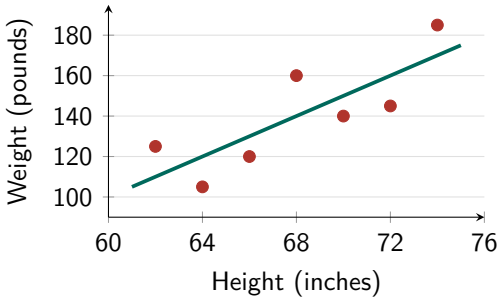
- Seven people's heights and weights, what is the best-fit line between weight (Y) and height (X)?

Person	Height (inches)	Weight (pounds)
1	62	125
2	64	105
3	66	120
4	68	160
5	70	140
6	72	145
7	74	185

Example, Height and Weight

- Seven people's heights and weights, what is the best-fit line between weight (Y) and height (X)?

Person	Height (inches)	Weight (pounds)
1	62	125
2	64	105
3	66	120
4	68	160
5	70	140
6	72	145
7	74	185



- Solution, the best-fit line is Weight $= -200 + 5 \times$ Height, so $\alpha = -200$ and $\beta = 5$

Estimating Beta by Regression

- Beta is a relationship between the stock's and the market's excess returns

Estimating Beta by Regression

- Beta is a relationship between the stock's and the market's excess returns
- Then how do we estimate beta? We learn from history

Estimating Beta by Regression

- Beta is a relationship between the stock's and the market's excess returns
- Then how do we estimate beta? We learn from history
- Run a linear regression of the security's excess return (Y) on the market portfolio's excess return (X),

$$R_{i,t} - r_{f,t} = \alpha + \beta \times (R_{\text{mkt},t} - r_{f,t}) + e_{i,t}$$

  - $R_{i,t}$ is the return of stock i over period t, $r_{f,t}$ the risk-free rate, and $R_{\text{mkt},t}$ the market return over period t

Estimating Beta by Regression

- Beta is a relationship between the stock's and the market's excess returns
- Then how do we estimate beta? We learn from history
- Run a linear regression of the security's excess return (Y) on the market portfolio's excess return (X),

$$R_{i,t} - r_{f,t} = \alpha + \beta \times (R_{\text{mkt},t} - r_{f,t}) + e_{i,t}$$

  - $R_{i,t}$ is the return of stock i over period t, $r_{f,t}$ the risk-free rate, and $R_{\text{mkt},t}$ the market return over period t

- The coefficient β from this regression is the estimate of the stock's beta

Example, Measuring Beta

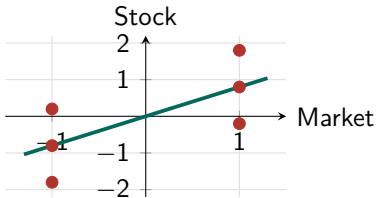
- A company's stock has the following excess returns, alongside the market portfolio's (both in %). What is its beta?

Month	Market excess return	Stock excess return
1	$+1$	$+0.8$
2	$+1$	$+1.8$
3	$+1$	-0.2
4	-1	-1.8
5	-1	$+0.2$
6	-1	-0.8

Example, Measuring Beta

- A company's stock has the following excess returns, alongside the market portfolio's (both in %). What is its beta?

Month	Market excess return	Stock excess return
1	$+1$	$+0.8$
2	$+1$	$+1.8$
3	$+1$	-0.2
4	-1	-1.8
5	-1	$+0.2$
6	-1	-0.8

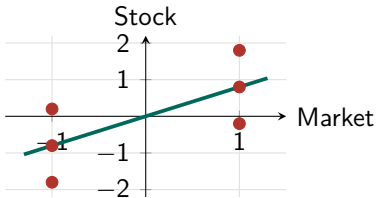


- Solution, the best-fit line is $\text{Stock} = 0 + 0.8 \times \text{Market}$

Example, Measuring Beta

- A company's stock has the following excess returns, alongside the market portfolio's (both in %). What is its beta?

Month	Market excess return	Stock excess return
1	$+1$	$+0.8$
2	$+1$	$+1.8$
3	$+1$	-0.2
4	-1	-1.8
5	-1	$+0.2$
6	-1	-0.8



- Solution, the best-fit line is Stock $= 0 + 0.8 \times$ Market
- The slope is the estimate of the stock's beta, here $\beta = 0.8$

Recap of Part 3

- **Beta**, how much a stock's return moves when the market's moves

Recap of Part 3

- **Beta**, how much a stock's return moves when the market's moves
- Cyclical businesses have high betas, producers of essentials low betas

Recap of Part 3

- **Beta**, how much a stock's return moves when the market's moves
- Cyclical businesses have high betas, producers of essentials low betas
- **Portfolio beta**, the weighted average of the securities' betas

Recap of Part 3

- **Beta**, how much a stock's return moves when the market's moves
- Cyclical businesses have high betas, producers of essentials low betas
- **Portfolio beta**, the weighted average of the securities' betas
- Estimated by regression, the slope of the stock's excess return on the market's

Module 8, Part 4

The CAPM

CAPM, the Relation between Risk (Beta) and Expected Return

- **Part 1**, risky securities earn a risk premium

CAPM, the Relation between Risk (Beta) and Expected Return

- **Part 1**, risky securities earn a risk premium
- **Parts 2 and 3**, the risk that matters is beta

CAPM, the Relation between Risk (Beta) and Expected Return

- **Part 1**, risky securities earn a risk premium
- **Parts 2 and 3**, the risk that matters is beta
- How much risk premium does a given beta earn?

CAPM, the Relation between Risk (Beta) and Expected Return

- **Part 1**, risky securities earn a risk premium
- **Parts 2 and 3**, the risk that matters is beta
- How much risk premium does a given beta earn?
- **Capital asset pricing model (CAPM)**, the expected return on any security is the risk-free rate plus its beta times the market risk premium,

$$E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$$

CAPM, the Relation between Risk (Beta) and Expected Return

- **Part 1**, risky securities earn a risk premium
- **Parts 2 and 3**, the risk that matters is beta
- How much risk premium does a given beta earn?
- **Capital asset pricing model (CAPM)**, the expected return on any security is the risk-free rate plus its beta times the market risk premium,

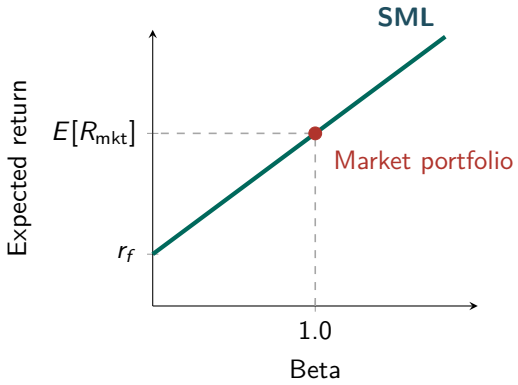
$$E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$$

  - $E[R_{\text{mkt}}] - r_f$, the **market risk premium**, the expected return on the market portfolio over the risk-free rate

CAPM, the Security Market Line

- **Security market line (SML)**, the graph of the relation between a stock's beta and its expected return,

$$E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$$

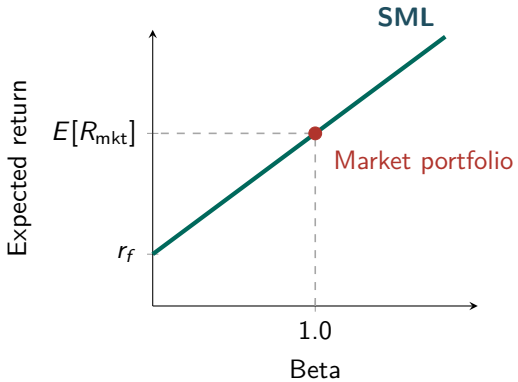


CAPM, the Security Market Line

- **Security market line (SML)**, the graph of the relation between a stock's beta and its expected return,

$$E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$$

- A straight line, it starts at r_f for $\beta = 0$ and passes through the market portfolio, $\beta = 1$ and $E[R_{\text{mkt}}]$



CAPM, Expected Returns for Selected Companies

- With $r_f = 3\%$ and a market risk premium $E[R_{\text{mkt}}] - r_f = 7\%$, the expected returns are,

Company	Business	Beta
Marathon Oil	Oil and gas	2.39
Disney	Entertainment	1.39
Intel	Chip maker	1.07
Coca-Cola	Beverages	0.70
Walmart	Retail	0.37

CAPM, Expected Returns for Selected Companies

- With $r_f = 3\%$ and a market risk premium $E[R_{\text{mkt}}] - r_f = 7\%$, the expected returns are,

Company	Business	Beta	$E[R]$ (%)
Marathon Oil	Oil and gas	2.39	19.7
Disney	Entertainment	1.39	12.7
Intel	Chip maker	1.07	10.5
Coca-Cola	Beverages	0.70	7.9
Walmart	Retail	0.37	5.6

Example, CAPM

- The expected return on the market portfolio is 10% and the risk-free rate is 3%
- What is the expected return on a stock with a beta of 0.5?

Example, CAPM

- The expected return on the market portfolio is 10% and the risk-free rate is 3%
- What is the expected return on a stock with a beta of 0.5?
- Solution, the market risk premium is $E[R_{\text{mkt}}] - r_f = 10\% - 3\% = 7\%$

Example, CAPM

- The expected return on the market portfolio is 10% and the risk-free rate is 3%
- What is the expected return on a stock with a beta of 0.5?
- Solution, the market risk premium is $E[R_{\text{mkt}}] - r_f = 10\% - 3\% = 7\%$

$$E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f) = 3\% + 0.5 \times 7\% = 6.5\%$$

Why CAPM?

- What is the idea behind the CAPM?

Why CAPM?

- What is the idea behind the CAPM?
- A savvy investor holds a diversified portfolio,

Why CAPM?

- What is the idea behind the CAPM?
- A savvy investor holds a diversified portfolio,

  $\rightarrow$ what matters is not a security's risk on its own but how much risk it adds to the portfolio

Why CAPM?

- What is the idea behind the CAPM?
- A savvy investor holds a diversified portfolio,
  - → what matters is not a security's risk on its own but how much risk it adds to the portfolio
  - → what matters is systematic risk, measured by beta, which diversification cannot remove

Why CAPM?

- What is the idea behind the CAPM?
- A savvy investor holds a diversified portfolio,
  - → what matters is not a security's risk on its own but how much risk it adds to the portfolio
  - → what matters is systematic risk, measured by beta, which diversification cannot remove
    - → Investors require a higher expected return to hold a high-beta security, as compensation for this risk

Why CAPM?

- What is the idea behind the CAPM?
- A savvy investor holds a diversified portfolio,
  - → what matters is not a security's risk on its own but how much risk it adds to the portfolio
  - → what matters is systematic risk, measured by beta, which diversification cannot remove
    - → Investors require a higher expected return to hold a high-beta security, as compensation for this risk
      They require nothing extra for firm-specific risk, because diversification removes it for free

Why CAPM?

- The formal proof, with all the math, is in William Sharpe's 1964 paper in the Journal of Finance



William F. Sharpe (Wikipedia)

Why CAPM?

- The formal proof, with all the math, is in William Sharpe's 1964 paper in the Journal of Finance
- It earned him the Nobel Prize in economics in 1990



William F. Sharpe (Wikipedia)

Practice Problem

- Risk and return, true or false? Explain or qualify as necessary

a. The capital asset pricing model predicts that a security with a beta of zero will provide an expected return of zero

Practice Problem

- Risk and return, true or false? Explain or qualify as necessary

a. The capital asset pricing model predicts that a security with a beta of zero will provide an expected return of zero

b. An investor who puts $10,000 in Treasury bills and $20,000 in the market portfolio will have a portfolio beta of 2.0

Practice Problem

- Risk and return, true or false? Explain or qualify as necessary

a. The capital asset pricing model predicts that a security with a beta of zero will provide an expected return of zero

b. An investor who puts $10,000 in Treasury bills and $20,000 in the market portfolio will have a portfolio beta of 2.0

c. Investors demand higher expected rates of return from stocks with returns that are highly exposed to macroeconomic changes

Practice Problem

a. False. If $\beta = 0$, the asset's expected return should equal the risk-free rate, not zero

Practice Problem

a. False. If $\beta = 0$, the asset's expected return should equal the risk-free rate, not zero

b. False. The portfolio is one third in Treasury bills and two thirds in the market, so its beta is $(1/3 \times 0) + (2/3 \times 1.0) = 2/3 \approx 0.67$

Practice Problem

a. False. If $\beta = 0$, the asset's expected return should equal the risk-free rate, not zero

b. False. The portfolio is one third in Treasury bills and two thirds in the market, so its beta is $(1/3 \times 0) + (2/3 \times 1.0) = 2/3 \approx 0.67$

c. True. The asset's beta is a function of its sensitivity to macroeconomic risks, among other factors

Recap of Part 4

- **CAPM**, $E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$, only systematic risk (beta) is rewarded

Recap of Part 4

- **CAPM**, $E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$, only systematic risk (beta) is rewarded
- **Security market line**, the graph of the relation between a stock's beta and its expected return

Module 8, Part 5

From Expected Return to Discount Rate

From Expected Stock Returns to the Discount Rate

- In **Modules 6 and 7**, we discounted a project's cash flows at the **opportunity cost**, the return you could earn on a different project (or a financial asset) with similar risk

From Expected Stock Returns to the Discount Rate

- In **Modules 6 and 7**, we discounted a project's cash flows at the **opportunity cost**, the return you could earn on a different project (or a financial asset) with similar risk
- Two questions were left open, what is risk, and how is the opportunity cost related to risk?

From Expected Stock Returns to the Discount Rate

- In **Modules 6 and 7**, we discounted a project's cash flows at the **opportunity cost**, the return you could earn on a different project (or a financial asset) with similar risk
- Two questions were left open, what is risk, and how is the opportunity cost related to risk?
- Now we can answer both
  - The risk that matters is beta, the rest is diversified away (**Parts 2 and 3**)

From Expected Stock Returns to the Discount Rate

- In **Modules 6 and 7**, we discounted a project's cash flows at the **opportunity cost**, the return you could earn on a different project (or a financial asset) with similar risk
- Two questions were left open, what is risk, and how is the opportunity cost related to risk?
- Now we can answer both
  - The risk that matters is beta, the rest is diversified away (**Parts 2 and 3**)
  - For any beta, the CAPM gives the return investors can expect to earn on stocks with that beta, $r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$

From Expected Stock Returns to the Discount Rate

- In **Modules 6 and 7**, we discounted a project's cash flows at the **opportunity cost**, the return you could earn on a different project (or a financial asset) with similar risk
- Two questions were left open, what is risk, and how is the opportunity cost related to risk?
- Now we can answer both
  - The risk that matters is beta, the rest is diversified away (**Parts 2 and 3**)
  - For any beta, the CAPM gives the return investors can expect to earn on stocks with that beta, $r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$
- Is this return the project's discount rate r in capital budgeting?

Assumption, No Debt

- Throughout this part, assume every company is financed by shareholders alone, with no debt

Assumption, No Debt

- Throughout this part, assume every company is financed by shareholders alone, with no debt
- Then a stock's expected return is the return the company's investors require

Assumption, No Debt

- Throughout this part, assume every company is financed by shareholders alone, with no debt
- Then a stock's expected return is the return the company's investors require
- And a stock's beta is the beta of the company's business

Assumption, No Debt

- Throughout this part, assume every company is financed by shareholders alone, with no debt
- Then a stock's expected return is the return the company's investors require
- And a stock's beta is the beta of the company's business
- Most companies do borrow, **Module 9** adds debt

Why Is the Expected Return on Same-Beta Stocks the Discount Rate?

- A company invests its shareholders' money. Consider a project with beta β

Why Is the Expected Return on Same-Beta Stocks the Discount Rate?

- A company invests its shareholders' money. Consider a project with beta β
- The shareholders could instead take the money and buy stocks with the same beta β themselves. By the CAPM, they would expect to earn $r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$

Why Is the Expected Return on Same-Beta Stocks the Discount Rate?

- A company invests its shareholders' money. Consider a project with beta β
- The shareholders could instead take the money and buy stocks with the same beta β themselves. By the CAPM, they would expect to earn $r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$
- So the project must be expected to earn at least that. If it earns less, the shareholders would rather have the money and buy the stocks

Why Is the Expected Return on Same-Beta Stocks the Discount Rate?

- A company invests its shareholders' money. Consider a project with beta β
- The shareholders could instead take the money and buy stocks with the same beta β themselves. By the CAPM, they would expect to earn $r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$
- So the project must be expected to earn at least that. If it earns less, the shareholders would rather have the money and buy the stocks
- Discounting at $r = r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$ checks exactly this, NPV > 0 means the project beats the same-beta stocks

Why Is the Expected Return on Same-Beta Stocks the Discount Rate?

- A company invests its shareholders' money. Consider a project with beta β
- The shareholders could instead take the money and buy stocks with the same beta β themselves. By the CAPM, they would expect to earn $r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$
- So the project must be expected to earn at least that. If it earns less, the shareholders would rather have the money and buy the stocks
- Discounting at $r = r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$ checks exactly this, NPV > 0 means the project beats the same-beta stocks
- So this return is the project's **opportunity cost of capital**, the discount rate r in capital budgeting

Why Is the Expected Return on Same-Beta Stocks the Discount Rate?

- A company invests its shareholders' money. Consider a project with beta β
- The shareholders could instead take the money and buy stocks with the same beta β themselves. By the CAPM, they would expect to earn $r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$
- So the project must be expected to earn at least that. If it earns less, the shareholders would rather have the money and buy the stocks
- Discounting at $r = r_f + \beta \times (E[R_{\text{mkt}}] - r_f)$ checks exactly this, NPV > 0 means the project beats the same-beta stocks
- So this return is the project's **opportunity cost of capital**, the discount rate r in capital budgeting

Discount rate of a project = Expected return on stocks with the same beta

Capital Budgeting with the CAPM, Three Steps

1. Estimate the beta of the project

Capital Budgeting with the CAPM, Three Steps

1. Estimate the beta of the project
2. Plug it into the CAPM to get the discount rate,

$$r = r_f + \beta_{\text{project}} \times (E[R_{\text{mkt}}] - r_f)$$

Capital Budgeting with the CAPM, Three Steps

1. Estimate the beta of the project
2. Plug it into the CAPM to get the discount rate,
$$r = r_f + \beta_{\text{project}} \times (E[R_{\text{mkt}}] - r_f)$$
3. Discount the project's cash flows at r as in **Module 7**, accept if NPV > 0 (equivalently, if IRR $> r$)

Example, the Project's Beta Sets the Discount Rate

- A project requires a $100 million investment today and pays a one-time cash flow with expected amount $105 million one year later
- The risk-free rate is 1% and the market risk premium is 7%

Example, the Project's Beta Sets the Discount Rate

- A project requires a \$100 million investment today and pays a one-time cash flow with expected amount \$105 million one year later
- The risk-free rate is 1% and the market risk premium is 7%
- Should the company invest if the future cash flow is riskless ($\beta = 0$)?

Example, the Project's Beta Sets the Discount Rate

- A project requires a \$100 million investment today and pays a one-time cash flow with expected amount \$105 million one year later
- The risk-free rate is 1% and the market risk premium is 7%
- Should the company invest if the future cash flow is riskless ($\beta = 0$)?
- Should the company invest if the future cash flow has the same risk as the market portfolio ($\beta = 1$)?

Example, the Project's Beta Sets the Discount Rate

- Solution, compare the project with what the same \$100 million would earn in securities with the same beta (\$ millions),

Beta	Same-beta securities	Discount rate r	\$100 becomes	Project pays	Decision

Example, the Project's Beta Sets the Discount Rate

- Solution, compare the project with what the same \$100 million would earn in securities with the same beta (\$ millions),

Beta	Same-beta securities	Discount rate r	\$100 becomes	Project pays	Decision
0	Treasury bills	$1\% + 0 \times 7\% = 1\%$	101	105	**Accept**

- $\beta = 0$, Treasury bills turn \$100 million into \$101 million, the project into \$105 million, more. NPV $= -100 + 105/1.01 = 3.96 > 0$, accept

Example, the Project's Beta Sets the Discount Rate

- Solution, compare the project with what the same \$100 million would earn in securities with the same beta (\$ millions),

Beta	Same-beta securities	Discount rate r	\$100 becomes	Project pays	Decision
0	Treasury bills	$1\% + 0 \times 7\% = 1\%$	101	105	**Accept**
1	The market portfolio	$1\% + 1 \times 7\% = 8\%$	108	105	**Reject**

- $\beta = 0$, Treasury bills turn \$100 million into \$101 million, the project into \$105 million, more. NPV $= -100 + 105/1.01 = 3.96 > 0$, accept
- $\beta = 1$, the market portfolio turns \$100 million into \$108 million, the project into \$105 million, less. NPV $= -100 + 105/1.08 = -2.78 < 0$, reject

Example, the Project's Beta Sets the Discount Rate

- Solution, compare the project with what the same \$100 million would earn in securities with the same beta (\$ millions),

Beta	Same-beta securities	Discount rate r	\$100 becomes	Project pays	Decision
0	Treasury bills	$1\% + 0 \times 7\% = 1\%$	101	105	**Accept**
1	The market portfolio	$1\% + 1 \times 7\% = 8\%$	108	105	**Reject**

- $\beta = 0$, Treasury bills turn \$100 million into \$101 million, the project into \$105 million, more. NPV $= -100 + 105/1.01 = 3.96 > 0$, accept
- $\beta = 1$, the market portfolio turns \$100 million into \$108 million, the project into \$105 million, less. NPV $= -100 + 105/1.08 = -2.78 < 0$, reject
- Same cash flows, opposite answers. The discount rate comes from the project's beta, and accepting only the projects with NPV > 0 at this rate makes the shareholders better off

Example, a Soup Company Drills for Oil

- A company makes soup, a producer of essentials with a beta of 0.5

Example, a Soup Company Drills for Oil

- A company makes soup, a producer of essentials with a beta of 0.5
- Recall the assumption, no debt, so its stock's beta is the beta of its business

Example, a Soup Company Drills for Oil

- A company makes soup, a producer of essentials with a beta of 0.5
- Recall the assumption, no debt, so its stock's beta is the beta of its business
- With $r_f = 3\%$ and a market risk premium of 7%, the expected return on its own stock is $3\% + 0.5 \times 7\% = 6.5\%$

Example, a Soup Company Drills for Oil

- A company makes soup, a producer of essentials with a beta of 0.5
- Recall the assumption, no debt, so its stock's beta is the beta of its business
- With $r_f = 3\%$ and a market risk premium of 7%, the expected return on its own stock is $3\% + 0.5 \times 7\% = 6.5\%$
- Now suppose it decides to drill for oil, an oil and gas business with a beta of 2.0

Example, a Soup Company Drills for Oil

- A company makes soup, a producer of essentials with a beta of 0.5
- Recall the assumption, no debt, so its stock's beta is the beta of its business
- With $r_f = 3\%$ and a market risk premium of 7%, the expected return on its own stock is $3\% + 0.5 \times 7\% = 6.5\%$
- Now suppose it decides to drill for oil, an oil and gas business with a beta of 2.0
- The soup business is worth \$8 billion, the oil project would be worth \$2 billion

Example, a Soup Company Drills for Oil

- A company makes soup, a producer of essentials with a beta of 0.5
- Recall the assumption, no debt, so its stock's beta is the beta of its business
- With $r_f = 3\%$ and a market risk premium of 7%, the expected return on its own stock is $3\% + 0.5 \times 7\% = 6.5\%$
- Now suppose it decides to drill for oil, an oil and gas business with a beta of 2.0
- The soup business is worth \$8 billion, the oil project would be worth \$2 billion
- Should the oil project be discounted at the company's 6.5%?

Example, a Soup Company Drills for Oil

- The company is a portfolio of its two businesses, with weights 80% and 20%, so its beta is the weighted average (**Part 3**),

$$\beta_{\text{company}} = 0.8 \times \beta_{\text{soup}} + 0.2 \times \beta_{\text{oil}} = 0.8 \times 0.5 + 0.2 \times 2.0 = 0.8$$

Example, a Soup Company Drills for Oil

- The company is a portfolio of its two businesses, with weights 80% and 20%, so its beta is the weighted average (**Part 3**),

$$\beta_{\text{company}} = 0.8 \times \beta_{\text{soup}} + 0.2 \times \beta_{\text{oil}} = 0.8 \times 0.5 + 0.2 \times 2.0 = 0.8$$

- The oil project drags up the company's systematic risk, so its shareholders now require more on the company as a whole, 8.6% instead of 6.5%,

$$r_{\text{company}} = r_f + \beta_{\text{company}} \times (E[R_{\text{mkt}}] - r_f) = 3\% + 0.8 \times 7\% = 8.6\%$$

Example, a Soup Company Drills for Oil

- The company is a portfolio of its two businesses, with weights 80% and 20%, so its beta is the weighted average (**Part 3**),

$$\beta_{\text{company}} = 0.8 \times \beta_{\text{soup}} + 0.2 \times \beta_{\text{oil}} = 0.8 \times 0.5 + 0.2 \times 2.0 = 0.8$$

- The oil project drags up the company's systematic risk, so its shareholders now require more on the company as a whole, 8.6% instead of 6.5%,

$$r_{\text{company}} = r_f + \beta_{\text{company}} \times (E[R_{\text{mkt}}] - r_f) = 3\% + 0.8 \times 7\% = 8.6\%$$

- That is exactly the weighted average of each business's own CAPM rate,

$$r_{\text{company}} = 0.8 \times r_{\text{soup}} + 0.2 \times r_{\text{oil}} = 0.8 \times 6.5\% + 0.2 \times 17\% = 8.6\%$$

Example, a Soup Company Drills for Oil

- Suppose the oil project earns only 6.5%, "clearing" the old company hurdle. Then the company earns

$$0.8 \times 6.5\% + 0.2 \times 6.5\% = 6.5\% < 8.6\%$$

Example, a Soup Company Drills for Oil

- Suppose the oil project earns only 6.5%, "clearing" the old company hurdle. Then the company earns

$$0.8 \times 6.5\% + 0.2 \times 6.5\% = 6.5\% < 8.6\%$$

- The shareholders are underpaid for the systematic risk they now carry, the project destroyed value

Example, a Soup Company Drills for Oil

- Suppose the oil project earns only 6.5%, "clearing" the old company hurdle. Then the company earns

$$0.8 \times 6.5\% + 0.2 \times 6.5\% = 6.5\% < 8.6\%$$

- The shareholders are underpaid for the systematic risk they now carry, the project destroyed value
- For the company to earn its 8.6%, the oil project must earn the rate for oil and gas businesses,

$$r_{\text{oil}} = r_f + \beta_{\text{oil}} \times (E[R_{\text{mkt}}] - r_f) = 3\% + 2.0 \times 7\% = 17\%$$

Example, a Soup Company Drills for Oil

- Suppose the oil project earns only 6.5%, "clearing" the old company hurdle. Then the company earns

$$0.8 \times 6.5\% + 0.2 \times 6.5\% = 6.5\% < 8.6\%$$

- The shareholders are underpaid for the systematic risk they now carry, the project destroyed value
- For the company to earn its 8.6%, the oil project must earn the rate for oil and gas businesses,

$$r_{\text{oil}} = r_f + \beta_{\text{oil}} \times (E[R_{\text{mkt}}] - r_f) = 3\% + 2.0 \times 7\% = 17\%$$

- So discount the oil project at 17%, its own rate, not at the company's 6.5%

The Project's Beta, Not the Company's

- A project's cost of capital depends on the systematic risk (beta) of the *project*, not on the risk of the company doing it

The Project's Beta, Not the Company's

- A project's cost of capital depends on the systematic risk (beta) of the *project*, not on the risk of the company doing it
- Never discount a project at the expected return on the company's own stock, unless the project has the same beta as the company's existing business

The Project's Beta, Not the Company's

- A project's cost of capital depends on the systematic risk (beta) of the *project*, not on the risk of the company doing it
- Never discount a project at the expected return on the company's own stock, unless the project has the same beta as the company's existing business
- Often measured with the beta of other companies exclusively in the same business as the project (the "twin method")

The Project's Beta, Not the Company's

- A project's cost of capital depends on the systematic risk (beta) of the *project*, not on the risk of the company doing it
- Never discount a project at the expected return on the company's own stock, unless the project has the same beta as the company's existing business
- Often measured with the beta of other companies exclusively in the same business as the project (the "twin method")
- A company can set two or more discount rates for different types of projects

Example, Capital Budgeting and Project Risk

- A company considers a project. It requires an investment of $350,000 today and brings in $100,000 a year for 5 years
- The risk-free rate is 4% and the market risk premium is 8%

Example, Capital Budgeting and Project Risk

- A company considers a project. It requires an investment of $350,000 today and brings in $100,000 a year for 5 years
- The risk-free rate is 4% and the market risk premium is 8%

1. What is the NPV of the project? Assume the project has the same risk as the company, which has a beta of 1.2

Example, Capital Budgeting and Project Risk

- A company considers a project. It requires an investment of $350,000 today and brings in $100,000 a year for 5 years
- The risk-free rate is 4% and the market risk premium is 8%

1. What is the NPV of the project? Assume the project has the same risk as the company, which has a beta of 1.2
2. What if the project is less risky than the company as a whole, with a beta of 1?

Example, Capital Budgeting and Project Risk

- A company considers a project. It requires an investment of $350,000 today and brings in $100,000 a year for 5 years
- The risk-free rate is 4% and the market risk premium is 8%

1. What is the NPV of the project? Assume the project has the same risk as the company, which has a beta of 1.2
2. What if the project is less risky than the company as a whole, with a beta of 1?

- Solution, the cash flows are a 5-year annuity of $100,000,

$$\text{NPV} = 100{,}000 \times \frac{1}{r}\left(1 - \frac{1}{(1+r)^5}\right) - 350{,}000$$

Example, Capital Budgeting and Project Risk

- A company considers a project. It requires an investment of $350,000 today and brings in $100,000 a year for 5 years
- The risk-free rate is 4% and the market risk premium is 8%

1. What is the NPV of the project? Assume the project has the same risk as the company, which has a beta of 1.2
2. What if the project is less risky than the company as a whole, with a beta of 1?

- Solution, the cash flows are a 5-year annuity of $100,000,

$$\text{NPV} = 100{,}000 \times \frac{1}{r}\left(1 - \frac{1}{(1+r)^5}\right) - 350{,}000$$

- 1. At average risk ($\beta = 1.2$), $r = 4\% + 1.2 \times 8\% = 13.6\%$, so NPV $= -\$3{,}366$

Example, Capital Budgeting and Project Risk

- A company considers a project. It requires an investment of $350,000 today and brings in $100,000 a year for 5 years
- The risk-free rate is 4% and the market risk premium is 8%

1. What is the NPV of the project? Assume the project has the same risk as the company, which has a beta of 1.2
2. What if the project is less risky than the company as a whole, with a beta of 1?

- Solution, the cash flows are a 5-year annuity of $100,000,

$$\text{NPV} = 100{,}000 \times \frac{1}{r}\left(1 - \frac{1}{(1+r)^5}\right) - 350{,}000$$

- 1. At average risk ($\beta = 1.2$), $r = 4\% + 1.2 \times 8\% = 13.6\%$, so NPV $= -\$3{,}366$
- 2. With $\beta = 1$, $r = 4\% + 1 \times 8\% = 12\%$, so NPV $= \$10{,}478$

Recap of Part 5

- Assumed no debt, every company financed by shareholders alone

Recap of Part 5

- Assumed no debt, every company financed by shareholders alone
- Discount rate of a project = Expected return on stocks with the same beta, the shareholders could earn that on their own

Recap of Part 5

- Assumed no debt, every company financed by shareholders alone
- Discount rate of a project = Expected return on stocks with the same beta, the shareholders could earn that on their own
- The expected return on the company's own stock is the right rate only for its same-beta projects

Recap of Part 5

- Assumed no debt, every company financed by shareholders alone
- Discount rate of a project = Expected return on stocks with the same beta, the shareholders could earn that on their own
- The expected return on the company's own stock is the right rate only for its same-beta projects
- Use the project's own beta (the twin method), not the company's

Key Takeaways

- $E[R_i] = r_f + \text{Risk premium}_i$, the risk premium of stocks is about 8.5%

Key Takeaways

- $E[R_i] = r_f + \text{Risk premium}_i$, the risk premium of stocks is about 8.5%
- Diversification removes firm-specific risk, not systematic risk, so beta is what matters

Key Takeaways

- $E[R_i] = r_f + \text{Risk premium}_i$, the risk premium of stocks is about 8.5%
- Diversification removes firm-specific risk, not systematic risk, so beta is what matters
- **CAPM**, $E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$, the risk premium a given beta earns

Key Takeaways

- $E[R_i] = r_f + \text{Risk premium}_i$, the risk premium of stocks is about 8.5%
- Diversification removes firm-specific risk, not systematic risk, so beta is what matters
- **CAPM**, $E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$, the risk premium a given beta earns
- Discount rate of a project = Expected return on stocks with the same beta

Preview of Module 9

- This module assumed companies financed by shareholders alone

Preview of Module 9

- This module assumed companies financed by shareholders alone
- Most companies also borrow, so their investors are lenders and shareholders

Preview of Module 9

- This module assumed companies financed by shareholders alone
- Most companies also borrow, so their investors are lenders and shareholders
- **Module 9**, the discount rate becomes the weighted average cost of capital (WACC), a mix of what lenders and shareholders require

Thank you!

Please let me know if you have any questions.